

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

A passive interface stops OSPF Hello packets and prevents neighbor adjacency on that interface, but the connected network can still be advertised through other active OSPF interfaces. Therefore, R3 can still learn R2's LAN route indirectly through R1, even though R2 and R3 are not direct OSPF neighbors.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65 = 64 (R3's serial link to R2) + 1 (R2's G0/0 LAN link)

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

This allows OSPF to calculate route costs more accurately for the real link speeds in modern networks, so that faster links like Gigabit Ethernet do not all end up with the same cost as Fast Ethernet.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24 = R1 S0/0/1 + R3 G0/0 = 781 + 1 = 782

The cost to 192.168.23.0/30 = R1 S0/0/1 + R3 S0/0/1 = 781 + 64 = 845

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562. Because each serial link now has OSPF cost 781 after all serial interfaces are set to bandwidth 128, and reaching the 192.168.23.0/24 network from R1 requires traveling two serial costs, 781 + 781 = 1562.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because the cost of the direct path from R1 to R3 increased to $R1-S0/0/1 + R3-G0/0 = 1565 + 1 = 1566$, while the cost of the alternative path through R2 is $R1-S0/0/0 + R2-S0/0/1 + R3-G0/0 = 781 + 781 + 1 = 1563$, so OSPF selects that lower-cost route instead.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The router ID should be controlled because it uniquely identifies the router in OSPF and affects DR/BDR election. If it is based on a physical interface, it may change when that interface goes down, which can disrupt OSPF operation. Using a loopback address or the router-id command keeps the router ID stable.

2. Why is the DR/BDR election process not a concern in this lab?

DR/BDR election is mainly relevant on multi-access broadcast networks with multiple OSPF routers on the same segment. In this lab, the router-to-router connections are serial point-to-point links, which do not use DR/BDR election, and each LAN has only one router participating in OSPF on that segment.

3. Why would you want to set an OSPF interface to passive?

Setting an OSPF interface to passive can reduce unnecessary OSPF Hello traffic to free up bandwidth, improve security by not forming adjacencies there, and still allow the connected network to be advertised into OSPF.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF is hierarchical because it is divided into areas, with area 0 acting as the backbone. This area-based design reduces routing overhead and supports scalable network growth.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route will be used even though it has the highest metric, because EIGRP has a lower administrative distance than OSPF and RIP.